agent_hw2 pdf

Implementation Logic

system prompt

```
 """You are a rigorous assistant. Use reasoning and tool-base
d search to produce precise answers.

STANDARDS:
- Prefer authoritative and recent sources.
- Cross-check important facts when possible.
- Reject conflicting or low-quality evidence.
- If data is insufficient, explain what is missing.

PROCESS:
1. Break the question into needed facts.
2. Choose a precise search query for each fact.
3. Compare results and compute if needed.
4. Give a concise, evidence-based conclusion.

RESPONSE FORMAT (required):
- Thought: your reasoning.
- Action: Search[your query] OR Action: Final Answer

FEW-SHOT EXAMPLE:
User: Which country has a higher GDP per capita, Japan or Sou
th Korea?

Thought: I need GDP and population for both countries.

Action: Search[Japan GDP and population]
```

```
Observation: Japan GDP is about 4.2 trillion USD, population
is about 125 million.

Thought: Now get South Korea's data.

Action: Search[South Korea GDP and population]

Observation: South Korea GDP is about 1.8 trillion USD, popul
ation is about 51 million.

Thought: Compute GDP per capita: Japan ≈ 33600, South Korea ≈
35294.

Action: Final Answer
Answer: South Korea has a higher GDP per capita than Japan.

For actual outputs in this system, the final step must be: Ac
tion: Final Answer"""
```

few shot example

這個few shot example 可以讓ai知道不要直接將整個問題丟上去查詢 而是先分別查詢兩個國家資料後 最後再進行統整輸出

```


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以下這個example展示了當面對action 回傳的資料無法正確判斷時 下次的reasoning 應該怎麼做

```
FEW-SHOT EXAMPLE:

User: Who is the CEO of the company that created the Claude A
I model?

Thought: I need to find which company created Claude first.

Action: Search[Claude AI CEO]

Observation: Results are unclear and do not directly mention
the CEO.

Thought: The query is too direct. I should first identify the
company behind Claude.

Action: Search[Claude AI company]
```

```
Observation: Claude is developed by a company called Anthropi
c.

Thought: Now I need the CEO of Anthropic.

Action: Search[Anthropic CEO]

Observation: The CEO of Anthropic is Dario Amodei.

Thought: I now have the CEO of the company that created Claud
e.

Action: Final Answer
Answer: Dario Amodei
```

Benchmark Traces

問題1

What fraction of Japan's population is Taiwan's population as of 2025?

可以看到他順利地將其拆解為 先找日本的人口 然後在下次的迴圈查詢台灣區人口

最後iteration 2 就認定資料已經足夠輸出 final answer 後再輸出自己的答案

```
Enter your query: What fraction of Japan's population is Taiwan's population as of 2025?

================================================================================
ITERATION 0
================================================================================
Thought: I need to find the population estimates for both Japan and Taiwan for 2025 to compute the fraction.

Action: Search[Japan population 2025]
[observation]
Japan Population reached 123.2 million people in Dec 2025, compared with the previously reported figure of 123.7 m
illion people in Dec 2024.

================================================================================
ITERATION 1
================================================================================
Thought: Now, I need to find the estimated population of Taiwan for 2025.

Action: Search[Taiwan population 2025]
[observation]
## Taiwan Population (1950-2025). Population Economy Trade Health Education Development Labor Force Environment Cr
ime Immigration Other. Population Growth Rate Density Urban Rural Life Expectancy Birth Rate Death Rate Infant Mor
tality Rate Fertility Rate. ##### Total current population for Taiwan in 2025 was **23,974,031**, a **0.1% increas
e** from 2024. * Total population for Taiwan in 2024 was **23,950,214**, a **0.11% increase** from 2023. * Total p
opulation for Taiwan in 2023 was **23,923,276**, a **0.13% increase** from 2022. * Total population for Taiwan in
2022 was **23,893,394**, a **0.14% increase** from 2021. Total population is based on the de facto definition of p
opulation, which counts all residents regardless of legal status or citizenship. The values shown are midyear esti
mates. The full historical dataset is available for download here: Population of Taiwan | Historical Data | 1950 -
 2025.

================================================================================
ITERATION 2
================================================================================
Thought: I have the populations for both Japan (123.2 million) and Taiwan (23,974,031) for 2025. Now, I will compu
te the fraction of Taiwan's population relative to Japan's population.

[FINAL ANSWER]
================================================================================
Answer: Taiwan's population is approximately 19.45% of Japan's population in 2025.
any other question? (type 'exit' to quit):
```

問題2

Compare the main display specs of iPhone 15 and Samsung S24.

一樣先查詢iphone 15 後查詢 samsung s24

並且在最後的final answer 有輸出兩種手機的尺寸和使用的螢幕類型跟刷新頻率

```
================================================================================
Answer: Taiwan's population is approximately 19.45% of Japan's population in 2025.
any other question? (type 'exit' to quit): Compare the main display specs of iPhone 15 and Samsung S24.

================================================================================
ITERATION 0
================================================================================
Thought: I need to gather the main display specifications for both the iPhone 15 and the Samsung Galaxy S24, inclu
ding screen size, resolution, refresh rate, and technology used.

Action: Search[iPhone 15 display specifications]
[observation]
# Apple iPhone 15. | Dimensions | 147.6 x 71.6 x 7.8 mm |. ### Screen and keys. | Screen type | XDR OLED touch scr
een, HDR10, Dolby Vision, 16 million colours |. | Screen size | 6.1 inches |. | Main camera | Dual 48 MP, f/1.6, 2
6mm (wide) + 12 MP, f/2.4, 13mm, 120° (ultra wide) |. | Front camera | Dual 12 MP, f/1.9, 23mm (wide) + SL 3D, (de
pth/biometrics sensor) |. | Video recorder | 4K@24/25/30/60fps, 1080p@25/30/60/120/240fps, HDR, Dolby Vision HDR (
up to 60fps), Cinematic mode (4K@30fps) |. | Chipset | Apple A16 Bionic (4 nm) |. | CPU | Hexa-core (2x3.46 GHz Ev
erest + 4x2.02 GHz Sawtooth) |. | GPU | Apple GPU (5-core graphics) |. | 3.5 mm jack | No |. ### Connectivity. | W
i-Fi | 802.11 a/b/g/n/ac/6, dual-band |. | Battery type | Non-removable Li-Ion 3349 mAh battery |. Excellent Very
Good Good Fair Low No Connection. Excellent Very Good Good Fair Low No Connection.

================================================================================
ITERATION 1
================================================================================
Thought: Now I need to find the display specifications for the Samsung Galaxy S24 to compare with the iPhone 15.

Action: Search[Samsung Galaxy S24 display specifications]
[observation]
Display. Size, 6.2-inch, 90.87% screen-to-body. Type, Dynamic AMOLED, 120Hz, HDR, 2600 nits. Resolution, 2340x1080
px, 19.5:9 ratio, 416 PPI.

================================================================================
ITERATION 2
================================================================================
Thought: I have the display specifications for both the iPhone 15 and Samsung Galaxy S24. Now I can summarize and
compare their main display specs.

[FINAL ANSWER]
================================================================================
Answer: The iPhone 15 has a 6.1-inch XDR OLED display, while the Samsung Galaxy S24 features a 6.2-inch Dynamic AM
OLED display with a 120Hz refresh rate.
```

問題3

因為老師原本的問題都會直接被我的ai one shot掉 可能是因為原本的問題太好找了 所以我把題目換成相較之下比較少見的arc browser 的 公司的ceo

可以看出第一次的查詢並沒有給出完整的結果(不知道 tavil 的問題?)

因此在iteration 2 的時候 agent 再次進行查詢

最後得到答案後輸出

```
Answer: Jaynti Kanani is the CEO of Morphic AI.
any other question? (type 'exit' to quit): Who is the CEO of the company behind Arc Browser?

================================================================================
ITERATION 0
================================================================================
Thought: I need to identify the company behind Arc Browser and find out who its CEO is.

Action: Search[Arc Browser company CEO]
[observation]
Co-founded The Browser Company to work with top talent on an ambitious, long-term project. Observed a shift to web
-based productivity tools and

================================================================================
ITERATION 1
================================================================================
Thought: I need to find the specific name of the CEO of The Browser Company, which developed the Arc Browser.

Action: Search[CEO of The Browser Company]
[observation]
## Competing with giants: An inside look at how The Browser Company builds product | Josh Miller (CEO). **Josh Mil
ler** is the CEO and co-founder of The Browser Company, where he helped build Arc, my go-to web browser. In today'
s episode, we get an inside look at the unique structure and values of The Browser Company and how their company c
ulture has helped them land some of the best talent in tech. Josh shares ways that his company embraces experiment
ation, including their "optimizing for feelings" approach to building, and explains why extreme transparency is at
 the forefront of everything they do. • Airbnb's Snow White project:. (19:38) How The Browser Company's values cre
ate a culture that allows them to ship so quickly. (45:57) Some of the odd teams at The Browser Company and why Jo
sh calls it a prototype-driven culture. (52:41) Why The Browser Company doesn't have traditional PMs. (1:02:55) Ho
w impactful moments in Josh's life influenced values at The Browser Company. (1:07:31) Why The Browser Company's a
pproach works for Arc.

================================================================================
ITERATION 2
================================================================================
Thought: I have confirmed that Josh Miller is the CEO of The Browser Company, which developed the Arc Browser.

[FINAL ANSWER]
================================================================================
Answer: Josh Miller is the CEO of The Browser Company, the company behind Arc Browser.
any other question? (type 'exit' to quit): 
```